

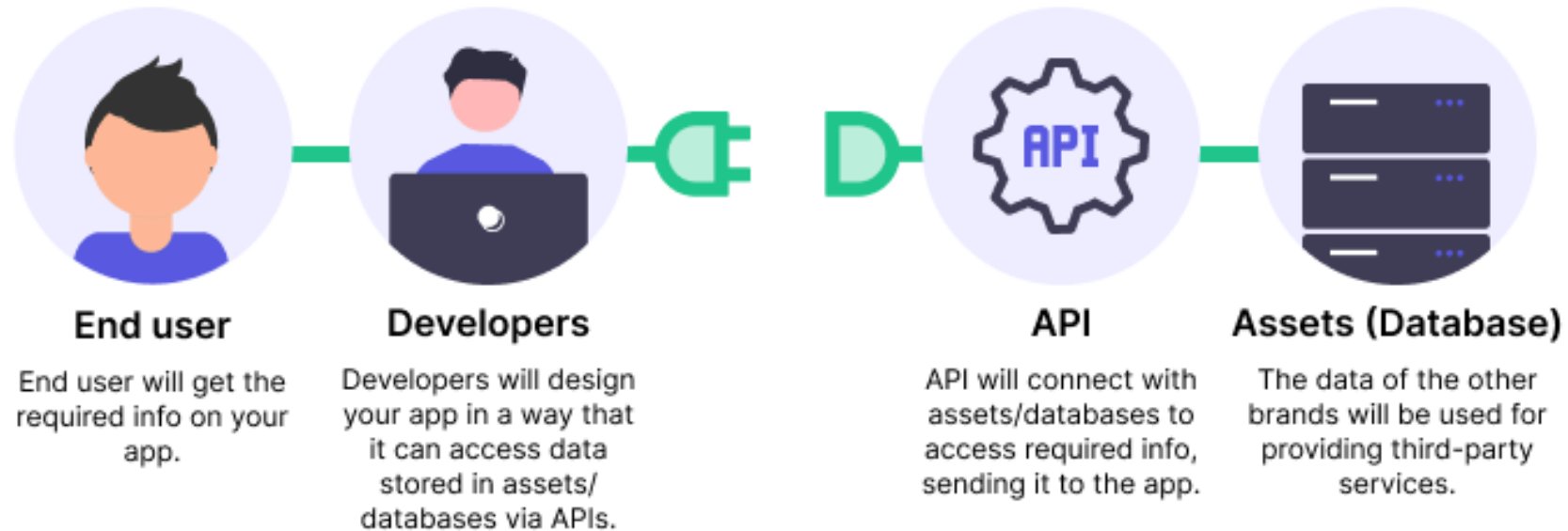
What is API

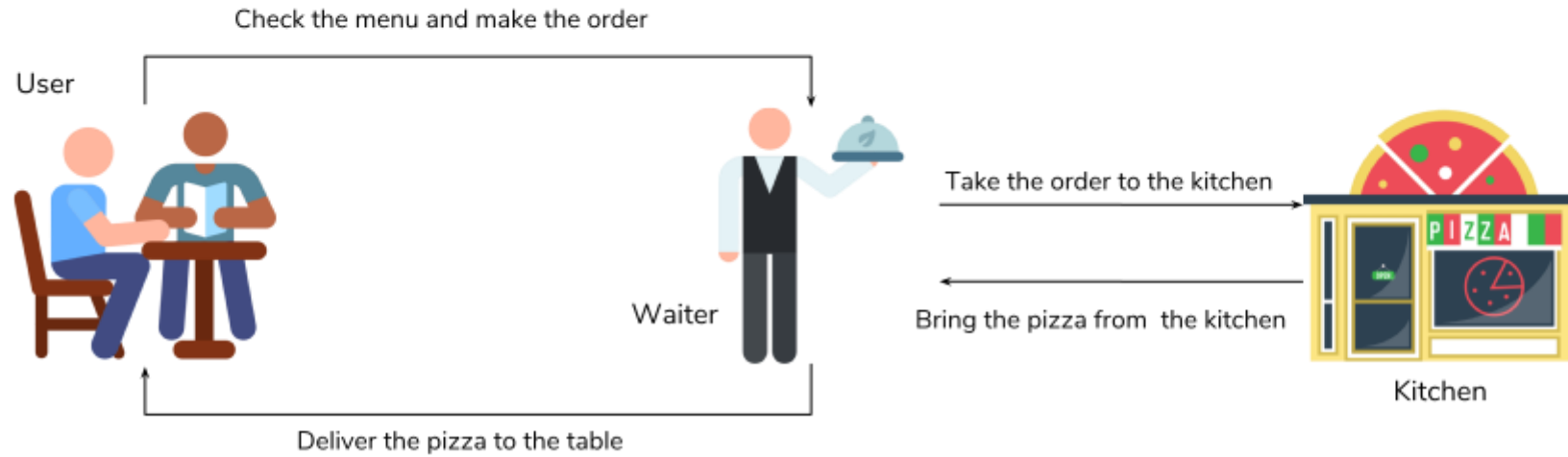
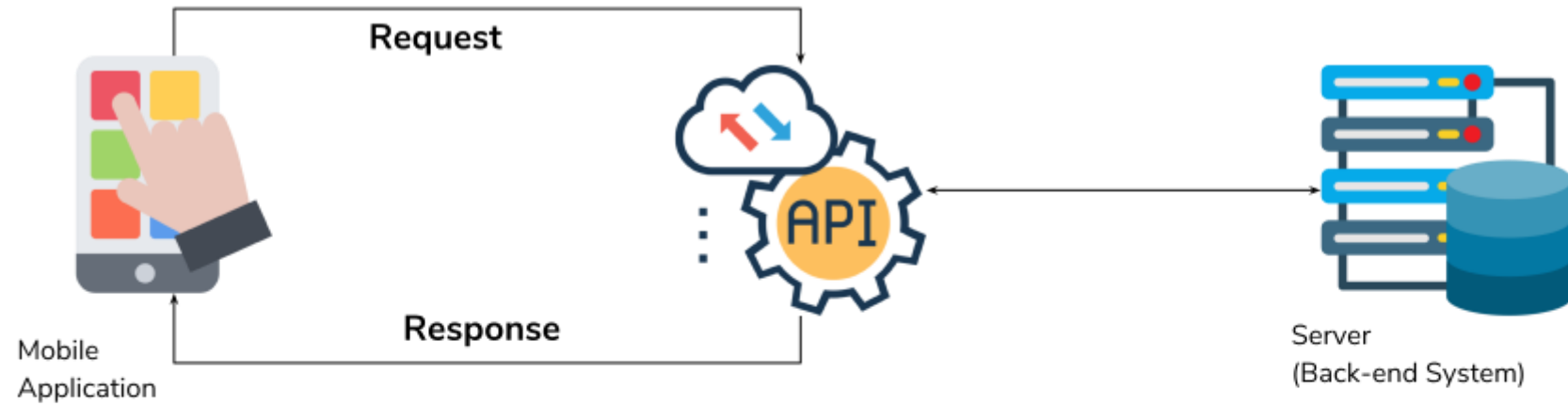
Application Programming Interface

- APIs are mechanisms that enable two software components to communicate with each other using a set of definitions and protocols.
- For example, the weather prediction software system contains daily weather data. The weather app on your phone “talks” to this system via APIs and shows you daily weather updates on your phone.



How does an API work?





Types of APIs



Private APIs

- ✦ Used to connect different software components within a single organization
- ✦ Not available for third-party use
- ✦ Some applications may include dozens or even hundreds of private APIs



Public APIs

- ✦ Provide public access to an organization's data, functionality, or services
- ✦ Can be integrated into third-party applications
- ✦ Some public APIs are available for free, while others are offered as billable products



Partner APIs

- ✦ Enable two or more companies to share data or functionality in order to collaborate
- ✦ Not available to the general public
- ✦ Leverage authentication mechanisms to restrict access

What is API level?

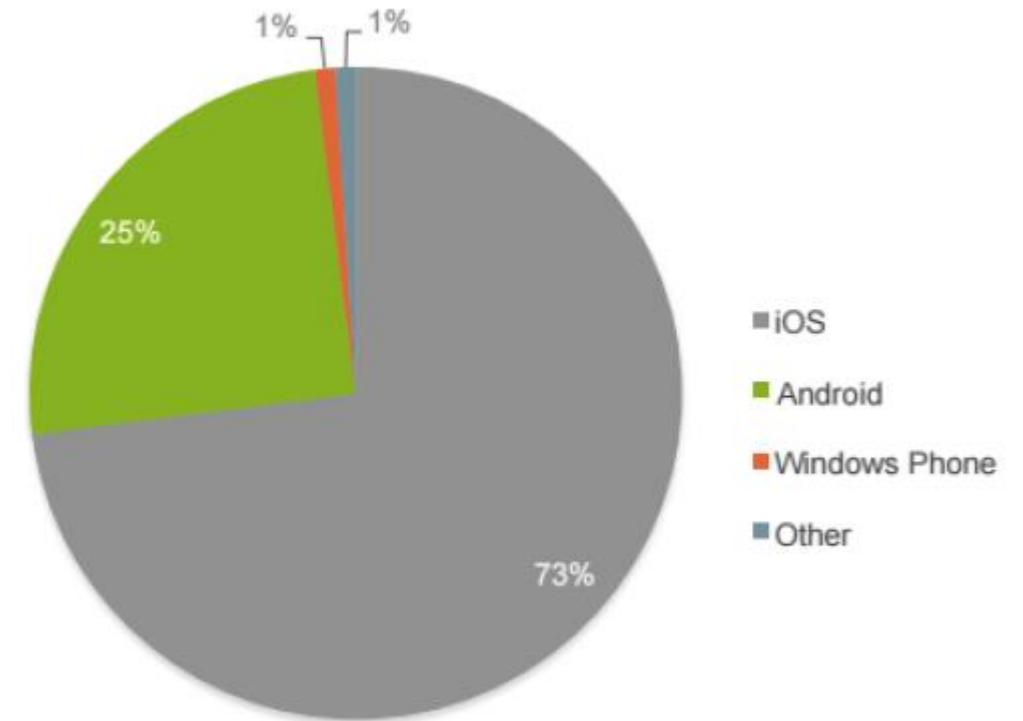
ANDROID PLATFORM VERSION	API LEVEL	CUMULATIVE DISTRIBUTION
4.0 Ice Cream Sandwich	15	
4.1 Jelly Bean	16	99.8%
4.2 Jelly Bean	17	99.2%
4.3 Jelly Bean	18	98.4%
4.4 KitKat	19	98.1%
5.0 Lollipop	21	94.1%
5.1 Lollipop	22	92.3%
6.0 Marshmallow	23	84.9%
7.0 Nougat	24	73.7%
7.1 Nougat	25	66.2%
8.0 Oreo	26	60.8%
8.1 Oreo	27	53.5%
9.0 Pie	28	39.5%
10. Android 10	29	8.2%

Android .. At a glance

- Mobile operating system based on **Linux kernel**.
- User Interface for touch screens.
- Used on over **80%** of all smartphones.
- Powers devices such as watches, TVs, and cars.
- Over 2 Million Android apps in Google Play store.
- Highly customizable for devices.
- Open source.

Android Popularity

- Developed by Google.
- Major contributors: Samsung, LG, Vodafone, Acer, Dell, HTC, Sony Ericsson, Intel, Wipro etc.
- Pre-installed Google apps:
 - Gmail,
 - Google Calendar
 - Google Play
 - Google Music
 - Chrome etc.
- Openness
- Customizable
- Affordable(low-cost)



The Android Studio

- Android Studio is the official integrated development environment (**IDE**) for Google's Android operating system, built on **JetBrains' IntelliJ IDEA software** and designed specifically for Android development.
- It is available for download on Windows, macOS and Linux based operating systems.

